BITS Pilani

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Introduction to Data Science

Data Wrangling (Contd.)

- *The slides presented here are obtained from the authors of the books and from various other contributors. I hereby acknowledge all the contributors for their material and inputs.*
- *We have added and modified slides to suit the requirements of the course.*

Data Similarity/Dissimilarity

Measuring Data Similarity and Dissimilarity

Various proximity measures

- Dissimilarity Matrix
- Proximity Measures for Nominal Attributes
- Proximity Measures for Ordinal Attributes
- Proximity Measures for Binary Attributes
    - Symmetric Binary Attributes
    - Asymmetric Binary Attributes
- Proximity Measures for Numeric Data
- Proximity Measures for Mixed Types
- Cosine Similarity

Similarity and Dissimilarity

- **Similarity**
  - Numerical measure of how alike two data objects are
  - Value is higher when objects are more alike
  - Often falls in the range [0,1]
- **Dissimilarity** (e.g., distance)
  - Numerical measure of how different two data objects are
  - Lower when objects are more alike
  - Minimum dissimilarity is often 0
  - Upper limit varies
- **Proximity** refers to a similarity or dissimilarity

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Data Matrix and Dissimilarity Matrix

- Data matrix
  - n data points with p dimensions
  - Two modes

$$\begin{bmatrix} x_{11} & \cdots & x_{1f} & \cdots & x_{1p} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ x_{i1} & \cdots & x_{if} & \cdots & x_{ip} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ x_{n1} & \cdots & x_{nf} & \cdots & x_{np} \end{bmatrix}$$

- Dissimilarity matrix
  - n data points, but registers only the distance
  - A triangular matrix
  - Single mode

$$\begin{bmatrix} 0 & & & & \\ d(2,1) & 0 & & & \\ d(3,1) & d(3,2) & 0 & & \\ \vdots & \vdots & \vdots & & \\ d(n,1) & d(n,2) & \cdots & \cdots & 0 \end{bmatrix}$$

Proximity Measure for Nominal Attributes

Categorical Attribute

- The similarity matrix considering only color attribute is shown below
- The dissimilarity matrix is as shown below:
  - 0 = identical; 1 = dissimilar
- m: # of matches, p: total # of attributes

Object	Color	Position	Size
1	R	L	L
2	B	C	M
3	G	C	M
4	R	L	H

	m	$\frac{p-m}{p}$		m	$\frac{p-m}{p}$		m	$\frac{p-m}{p}$
d(2,1)	0	$\frac{1-0}{1}=1$						
d(3,1)	0	$\frac{1-0}{1}=1$	d(3,2)	0	$\frac{1-0}{1}=1$			
d(4,1)	1	$\frac{1-1}{1}=0$	d(4,2)	0	$\frac{1-0}{1}=1$	d(4,3)	0	$\frac{1-0}{1}=1$

$$d = \begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \end{array} \begin{array}{c} \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\ \begin{bmatrix} 0 & & & \\ 1 & 0 & & \\ 1 & 1 & 0 & \\ 0 & 1 & 1 & 0 \end{bmatrix} \end{array}$$

Proximity Measure for Nominal Attributes

Categorical Attribute

- Dissimilarity matrix considering both the categorical attributes (i.e. color and position).
- The dissimilarity matrix is as shown below:
  - 0 = identical; 1 = dissimilar
- m: # of matches, p: total # of attributes

Object	Color	Position	Size
1	R	L	L
2	B	C	M
3	G	C	M
4	R	L	H
5	G	R	L
6	G	R	H

	m	$\frac{p-m}{p}$		m	$\frac{p-m}{p}$		m	$\frac{p-m}{p}$		m	$\frac{p-m}{p}$		m	$\frac{p-m}{p}$
d(2,1)	0	$\frac{2-0}{2}=1$												
d(3,1)	0	$\frac{2-0}{2}=1$	d(3,2)	1	$\frac{2-1}{2}=0.5$									
d(4,1)	2	$\frac{2-2}{2}=0$	d(4,2)	0	$\frac{2-0}{2}=1$	d(4,3)	0	$\frac{2-0}{2}=1$						
d(5,1)	0	$\frac{2-0}{2}=1$	d(5,2)	0	$\frac{2-0}{2}=1$	d(5,3)	1	$\frac{2-1}{2}=0.5$	d(5,4)	0	$\frac{2-0}{2}=1$			
d(6,1)	0	$\frac{2-0}{2}=1$	d(6,2)	0	$\frac{2-0}{2}=1$	d(6,3)	1	$\frac{2-1}{2}=0.5$	d(6,4)	0	$\frac{2-0}{2}=1$	d(6,5)	2	$\frac{2-2}{2}=0$

$$d = \begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \end{array} \begin{array}{c} \begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & 6 \end{array} \\ \begin{bmatrix} 0 & & & & & \\ 1 & 0 & & & & \\ 1 & 0.5 & 0 & & & \\ 0 & 1 & 1 & 0 & & \\ 1 & 1 & 0.5 & 1 & 0 & \\ 1 & 1 & 0.5 & 1 & 0 & 0 \end{bmatrix} \end{array}$$

Proximity Measure for Nominal Attributes

- Can take 2 or more states, e.g., red, yellow, blue, green (generalization of a binary attribute)
- Method 1: Simple matching
  - *m*: # of matches, *p*: total # of variables

$$d(i, j) = \frac{p - m}{p}$$

- Method 2: Use a large number of binary attributes
  - creating a new binary attribute for each of the *M* nominal states

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Ordinal Variables

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Proximity Measure For Ordinal Attributes

- Values of ordinal attribute follow a sequence (ordering)
  - E.g. Grade = {A, B, C}, where A >B >C
  - E.g., Size attribute: Small < Medium < Large < X-Large
- Suppose, *f* is an ordinal attribute and the set of values of
  - f = $\{a_1, a_2, \ldots, a_M\}$
  - *M* = the number of possible states that the ordinal attribute can have
- Let M values of *f* are ordered in ascending order as
  - $a_1 < a_2 < \cdots < a_M$
  - Let i^{th} attribute value a_i be ranked as i, i=1,2,.., M_f

Proximity Measure For Ordinal Attributes

- **Step 1)**
  - The value of f for the ith object is x_{if}, and f has M_f ordered states, representing the ranking $1, \ldots, M_f$
  - Replace each x_{if} by its corresponding rank, $r_{if} \in \{1, \ldots, M_f\}$
- **Step 2)**
  - Since each ordinal attribute can have a different number of states, it is often necessary to map the range of each attribute onto [0.0, 1.0] so that each attribute has equal weight
  - We perform such data normalization by replacing the rank r_{if} of the ith object in the fth attribute by

$$z_{if} = \frac{r_{if} - 1}{M_f - 1}$$

- **Step 3)**
  - Dissimilarity can then be computed using any of the distance measures used for numeric attributes, using z_{if} to represent the f value for the ith object

Proximity Measure For Ordinal Attributes

Ordinal Attribute

- Here, the attribute Test has three states:
  - Fair, Good and Excellent, so $M_f = 3$
- Step 1:
  - The four attribute values are assigned the ranks 3, 1, 2, and 3 respectively
- Step 2:
  - Normalizes the ranking by mapping ranks to [0.0, 1.0]
  - Norm(Rank 3) = $\frac{3-1}{3-1} = 1.0$; Norm(Rank 2) = $\frac{2-1}{3-1} = 0.5$; Norm(Rank 1) = $\frac{1-1}{3-1} = 0.0$
- Step 3:
  - Using Euclidean distance, a dissimilarity matrix is obtained as shown
- Therefore, students 1 and 2 are most dissimilar, as are students 2 and 4

$$\boxed{z_{if} = \frac{r_{if} - 1}{M_f - 1}}$$

Student	Test	Rank	Norm
1	Excellent	3	1.0
2	Fair	1	0.0
3	Good	2	0.5
4	Excellent	3	1.0

$$d = \begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \end{array} \begin{array}{c} \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\ \begin{bmatrix} 0 & & & \\ 1.0 & 0 & & \\ 0.5 & 0.5 & 0 & \\ 0 & 1.0 & 0.5 & 0 \end{bmatrix} \end{array}$$

	$\lvert a_i - a_j \rvert$		$\lvert a_i - a_j \rvert$		$\lvert a_i - a_j \rvert$
d(2,1)	$\lvert 1.0 - 0.0 \rvert = 1.0$				
d(3,1)	$\lvert 0.5 - 1.0 \rvert = 0.5$	d(3,2)	$\lvert 0.5 - 0.0 \rvert = 0.5$		
d(4,1)	$\lvert 1.0 - 1.0 \rvert = 0.0$	d(4,2)	$\lvert 1.0 - 0.0 \rvert = 1.0$	d(4,3)	$\lvert 1.0 - 0.5 \rvert = 0.5$

Proximity Measure For Ordinal Attributes

- Consider the following set of records, where each record is defined by two ordinal attributes
  - Size={Small, Medium, Large} such that S < M < L
  - Let's say
    - S = 1, M = 2, L = 3
  - Quality = {Excellent, Good, Fair, Poor} such that Ex > G > F > P
  - Let's say
    - Ex = 4, G = 3, F = 2, & P = 1

Object	Size	Quality
A	S (0.0)	G (0.67)
B	L (1.0)	Ex (1.0)
C	L (1.0)	P (0.0)
D	M (0.5)	F (0.33)

- Normalized values of Size {Small, Med, Large} = {0.0, 0.5, 1.0}:
  - Large= $\frac{3-1}{3-1} = 1$, Small= $\frac{1-1}{3-1} = 0$; Medium = $\frac{2-1}{3-1} = 0.5$;
- Normalized values of {Ex, G, F, P} = {1.0, 0.67, 0.33, 0.0}:
  - Ex = $\frac{4-1}{4-1} = 1.0$; G = $\frac{3-1}{4-1} = 0.67$; F = $\frac{2-1}{4-1} = 0.33$; P = $\frac{1-1}{4-1} = 0.0$

Proximity Measure For Ordinal Attributes

Object	Size	Quality
A	S (0.0)	G (0.67)
B	L (1.0)	Ex (1.0)
C	L (1.0)	P (0.0)
D	M (0.5)	F (0.33)

$$d = \begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \end{array} \begin{array}{c} \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\ \begin{bmatrix} 0 & & & \\ 1.053 & 0 & & \\ 1.204 & 1.0 & 0 & \\ 0.605 & 0.836 & 0.599 & 0 \end{bmatrix} \end{array}$$

	Distance Measure		Distance Measure
d(2,1)	$\sqrt{(1.0-0.0)^2+(1.0-0.67)^2} = \sqrt{1.0+0.1089} = \sqrt{1.1089} = 1.053$	d(3,1)	$\sqrt{(1.0-0.0)^2+(0.0-0.67)^2} = \sqrt{1.0+0.4489} = \sqrt{1.4489} = 1.204$
d(3,2)	$\sqrt{(1.0-1.0)^2+(0.0-1.0)^2} = \sqrt{0.0+1.0} = \sqrt{1.0} = 1.0$	d(4,1)	$\sqrt{(0.5-0.0)^2+(0.33-0.67)^2} = \sqrt{0.25+0.1156} = \sqrt{0.3656} = 0.605$
d(4,2)	$\sqrt{(0.5-1.0)^2+(0.33-1.0)^2} = \sqrt{0.25+0.4489} = \sqrt{0.6989} = 0.836$	d(4,3)	$\sqrt{(1.0-0.5)^2+(0.33-0.0)^2} = \sqrt{0.25+0.1089} = \sqrt{0.3589} = 0.599$

Ordinal Variables

- An ordinal variable can be discrete or continuous
- Order is important, e.g., rank
- Can be treated like interval-scaled
  - replace x_{if} by their rank $r_{if} \in \{1, \ldots, M_f\}$
  - map the range of each variable onto [0, 1] by replacing *i*-th object in the *f*-th variable by

$$z_{if} = \frac{r_{if} - 1}{M_f - 1}$$

  - compute the dissimilarity using methods for interval-scaled variables

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Proximity Measure For Binary Attributes

Proximity Measure For Binary Attributes

- A contingency table for binary data

		Object j		
		1	0	sum
Object i	1	q	r	$q+r$
	0	s	t	$s+t$
	sum	$q+s$	$r+t$	p

  - where q is the number of attributes that equal 1 for both objects i and j,
  - r is the number of attributes that equal 1 for object i but equal 0 for object j,
  - s is the number of attributes that equal 0 for object i but equal 1 for object j,
  - t is the number of attributes that equal 0 for both objects i and j.
  - The total number of attributes is p, where $p = q+r+s+t$.

Proximity Measure For Binary Attributes

Symmetric binary attributes

- For symmetric binary attributes, each state is equally valuable
- Dissimilarity that is based on symmetric binary attributes is called **symmetric binary dissimilarity**
- If objects *i* and *j* are described by symmetric binary attributes, then the dissimilarity between *i* and *j* is

Contingency Table for Binary Attributes

		Object *j*		
		1	0	sum
	1	q	r	$q+r$
Object *i*	0	s	t	$s+t$
	sum	$q+s$	$r+t$	p

$$d(i,j) = \frac{r+s}{q+r+s+t}$$

Proximity Measure For Binary Attributes

- Asymmetric binary attributes
  - For asymmetric binary attributes, the two states are not equally important
    - E.g., the *positive* (1) and *negative* (0) outcomes of a disease test
  - Given two asymmetric binary attributes, the agreement of two 1s (a positive match) is then considered more significant than that of two 0s (a negative match)
  - Therefore, such binary attributes are often considered "monary" (having one state)
  - The dissimilarity based on these attributes is called **asymmetric binary dissimilarity**
  - Here, the number of negative matches, *t*, is considered unimportant and is thus ignored in the following computation:

$$d(i,j) = \frac{r+s}{q+r+s}$$

Proximity Measure For Binary Attributes

- Measurement of similarity
  - We can measure the difference between two binary attributes based on the notion of similarity instead of dissimilarity
  - For example, the **asymmetric binary similarity** between the objects *i* and *j* can be computed as
  - $$\mathrm{sim}(i,j) = 1 - d(i,j) = 1 - \frac{r+s}{q+r+s} = \frac{q}{q+r+s}$$
  - The coefficient *sim(i, j) above* is called the **Jaccard coefficient**

Proximity Measure For Binary Attributes

Symmetric Binary Attributes

- Consider the following two dataset, where objects are defined with symmetric binary attributes.
  - Gender = {M, F}
  - Food = {V, N}
  - Caste = {H, M}
  - Education = {E, M}
  - Hobby = {T, C}
  - Job = {Y, N}

Object	Gender	Food	Caste	Education	Hobby	Job
Ahmed	M (1)	V (0)	M (1)	E (1)	C (1)	N (0)
Surekha	M (1)	N (1)	H (0)	M (0)	T (0)	N (0)
Mahesh	F (0)	N (1)	H (0)	E (1)	C (1)	Y (1)

$$d(Surekha, Ahmed) = \frac{r+s}{q+r+s+t} = \frac{1+3}{1+1+3+1}$$

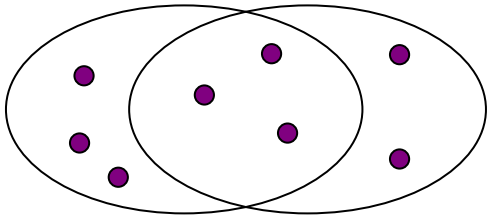
$$= \frac{4}{6} = 0.67$$

Continency Table (Ahmed & Surekha)

Object	Ahmed			
Surekha		1	0	Sum
	1	1	1	2
	0	3	1	4
	Sum	4	2	6

Jaccard Distance/Similarity

- For each application, we first need to define what "**distance**" means.
- Jaccard measures work with sets.
- **Jaccard distance/similarity**
  - The **Jaccard similarity** of two **sets** is the size of their intersection divided by the size of their union:
    $sim(C_1, C_2) = |C_1 \cap C_2| / |C_1 \cup C_2|$
  - **Jaccard distance:** $d(C_1, C_2) = 1 - |C_1 \cap C_2| / |C_1 \cup C_2|$



3 in intersection
8 in union
Jaccard similarity= 3/8
Jaccard distance = 5/8

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Proximity Measure For Binary Attributes

EXAMPLE

Suppose that a patient record table contains the attributes *name, gender, fever, cough, test-1, test-2, test-3*, and *test-4*, where *name* is an object identifier, *gender* is a symmetric attribute, and the remaining attributes are asymmetric binary. Compute the dissimilarity matrix for asymmetric binary attributes

Relational Table Where Patients Are Described by Binary Attributes

name	*gender*	*fever*	*cough*	*test-1*	*test-2*	*test-3*	*test-4*
Jack	M	Y	N	P	N	N	N
Jim	M	Y	Y	N	N	N	N
Mary	F	Y	N	P	N	P	N

Proximity Measure For Binary Attributes

EXAMPLE

name	gender	fever	cough	test-1	test-2	test-3	test-4
Jack	M	Y	N	P	N	N	N
Jim	M	Y	Y	N	N	N	N
Mary	F	Y	N	P	N	P	N

$$d(Jack, Jim) = \frac{1+1}{1+1+1} = 0.67,$$

$$d(Jack, Mary) = \frac{0+1}{2+0+1} = 0.33,$$

$$d(Jim, Mary) = \frac{1+2}{1+1+2} = 0.75.$$

Jack	Jim	
	1	0
1	q = 1	r = 1
0	s = 1	t = 3

Jack	Mary	
	1	0
1	q = 2	r = 0
0	s = 1	t = 3

Mary	Jim	
	1	0
1	q = 1	r = 1
0	s = 2	t = 2

Proximity Measure For Binary Attributes

- Distance measure for symmetric binary variables, dissimilarity:

$$d(i,j) = \frac{r+s}{q+r+s+t}.$$

- Distance measure for asymmetric binary variables, dissimilarity:

$$d(i,j) = \frac{r+s}{q+r+s}.$$

- Similarity between Asymmetric binary values is given by Jaccard coefficient:

$$sim(i,j) = \frac{q}{q+r+s} = 1 - d(i,j).$$

Proximity Measure For Numeric Attributes

Proximity Measure For Numeric Attributes

Dissimilarity of Numeric Data: Euclidean & Manhattan Dist.

- **Euclidean Distance** (i.e., straight line or "as the crow flies")
  - The Euclidean distance between objects *i* and *j* is defined as

$$d(i,j) = \sqrt{(x_{i1} - x_{j1})^2 + (x_{i2} - x_{j2})^2 + \cdots (x_{ip} - x_{jp})^2}$$

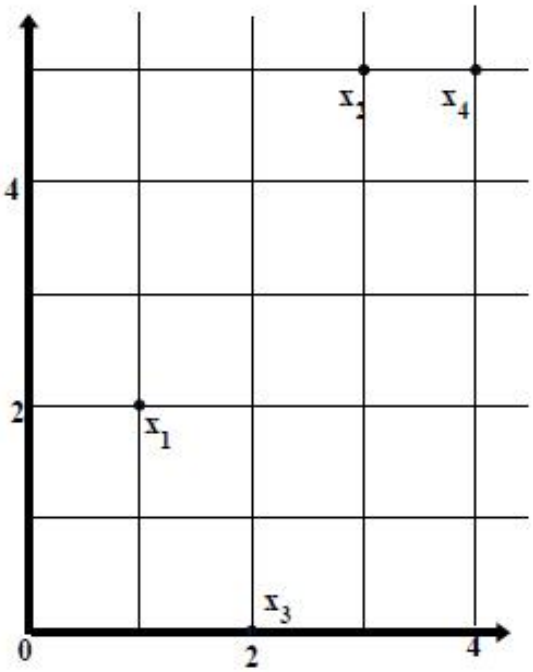
$$\begin{bmatrix} x_{11} & \cdots & x_{1f} & \cdots & x_{1p} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ x_{i1} & \cdots & x_{if} & \cdots & x_{ip} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ x_{n1} & \cdots & x_{nf} & \cdots & x_{np} \end{bmatrix}$$

- **Manhattan (or city block) distance**
  - It is the distance in blocks between any two points in a city (such as 2 blocks down and 3 blocks over for a total of 5 blocks)
  - The Manhattan distance between objects *i* and *j* is defined as

$$d(i,j) = |x_{i1} - x_{j1}| + |x_{i2} - x_{j2}| + \cdots |x_{ip} - x_{jp}|$$

Proximity Measure For Numeric Attributes

Euclidean Distance



point	attribute1	attribute2
x1	1	2
x2	3	5
x3	2	0
x4	4	5

Euclidean distance between x_1 & x_2

$$= \sqrt{(3-1)^2 + (5-2)^2}$$

$$= \sqrt{4+9} = \sqrt{13} = 3.61$$

Dissimilarity Matrix

(with Euclidean Distance)

	x1	*x2*	*x3*	*x4*
x1	0			
x2	3.61	0		
x3	5.1	5.1	0	
x4	4.24	1	5.39	0

Proximity Measure For Numeric Attributes

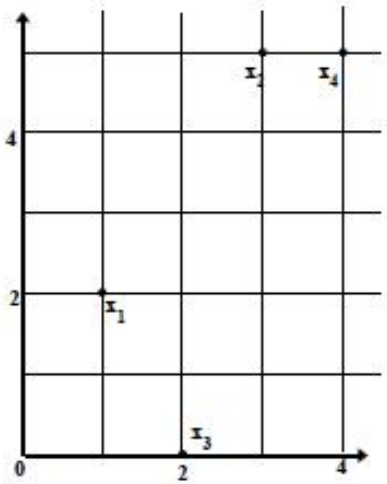
Manhattan Distance

Manhattan distance between x_1 & x_2

$$d(i,j) = |x_{i1} - x_{j1}| + |x_{i2} - x_{j2}| + \cdots |x_{ip} - x_{jp}|$$

$$d(x_2, x_1) = |3-1| + |5-2| = 2 + 3 = 5$$
$$d(x_3, x_1) = |2-1| + |0-2| = 1 + 2 = 3$$
$$d(x_3, x_2) = |2-3| + |0-5| = 1 + 5 = 6$$

point	attribute1	attribute2
x1	1	2
x2	3	5
x3	2	0
x4	4	5



Dissimilarity Matrix

L	x1	x2	x3	x4
x1	0			
x2	5	0		
x3	3	6	0	
x4	6	1	7	0

Proximity Measure For Numeric Attributes

Dissimilarity of Numeric Data: Minkowski Distance

- Minkowski distance is a generalization of the Euclidean and Manhattan distances
- It is defined as

$$d(i,j) = \sqrt[h]{|x_{i1} - x_{j1}|^h + |x_{i2} - x_{j2}|^h + \cdots + |x_{ip} - x_{jp}|^h}$$

- Where
  - $i = x_{i1}, x_{i2}, \ldots, x_{ip}$ and $j = x_{j1}, x_{j2}, \ldots, x_{jp}$ are two objects described by p numeric attributes
  - h is a real number such that $h \geq 1$
  - It is also called as L_h **norm**
  - When h = 1, it represents the Manhattan distance (i.e., L_1 norm)
  - When h = 2, it represents the Euclidean distance (i.e., L_2 norm)

Proximity Measure For Numeric Attributes

Dissimilarity of Numeric Data: Supremum Distance

- The **supremum distance** is a generalization of the Minkowski distance for h = ∞
  - also referred to as L_{max}, L_∞ **norm,** and as the **Chebyshev distance**
- To compute it, we find the attribute *f* that gives the maximum difference in values between the two objects
- This difference is the supremum distance, defined more formally as:

$$d(i,j) = \lim_{h\to\infty}\left(\sum_{f=1}^{p}|x_{i1}-x_{j1}|\right)^{1/h} = \max_f |x_{if}-x_{jf}|$$

  - The L^∞ norm is also known as the *uniform norm*

Proximity Measure For Numeric Attributes

Supremum

Supremum Distance between x_1 & x_2

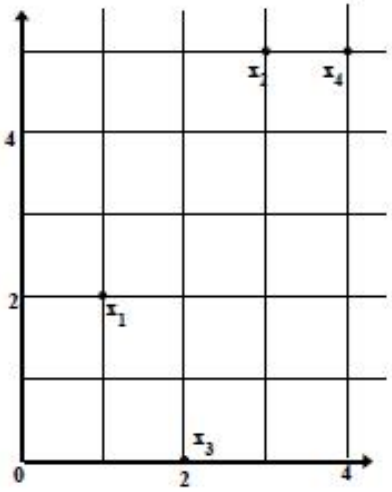
$$d(i,j) = \max_f |x_{if} - x_{jf}|$$

$$d(x_2, x_1) = Max\ (|3-1|, |5-2|) = 3$$

$$d(x_3, x_1) = Max\ (|2-1|, |0-2|) = 2$$

$$d(x_3, x_2) = Max\ (|2-3|, |0-5|) = 5$$

point	attribute1	attribute2
x1	1	2
x2	3	5
x3	2	0
x4	4	5



Dissimilarity Matrix

L_∞	x1	x2	x3	x4
x1	0			
x2	3	0		
x3	2	5	0	
x4	3	1	5	0

Proximity Measure For Complex Objects

Proximity Measure For Complex Objects

A data object (or record or tuple) may contain all attribute types

Nominal, symmetric binary, asymmetric binary, numeric, ordinal

One will have to combine their effects

Proximity Measure For Complex Objects

- Process all attribute types together, performing a single analysis
- Combine the different attributes into a single dissimilarity matrix, bringing all of the meaningful attributes onto a common scale of the interval [0.0, 1.0]
- Suppose that the data set contains *p* attributes of mixed type
- The dissimilarity *d(i, j)* between objects *i* and *j* is defined as:

$$d(i,j) = \frac{\sum_{f=1}^{p} \delta_{ij}^{(f)} d_{ij}^{(f)}}{\sum_{f=1}^{p} \delta_{ij}^{(f)}}$$

Attributes of Mixed Type

- A database may contain all attribute types
  - Nominal, symmetric binary, asymmetric binary, numeric, ordinal
- One may use a weighted formula to combine their effects

$$d(i,j) = \frac{\sum_{f=1}^{p} \delta_{ij}^{(f)} d_{ij}^{(f)}}{\sum_{f=1}^{p} \delta_{ij}^{(f)}}$$

  - f is binary or nominal:
    $d_{ij}^{(f)} = 0$ if $x_{if} = x_{jf}$, or $d_{ij}^{(f)} = 1$ otherwise
  - f is numeric: use the normalized distance
  - f is ordinal
    - Compute ranks r_{if} and
    - Treat z_{if} as interval-scaled

$$z_{if} = \frac{r_{if} - 1}{M_f - 1}$$

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Proximity Measure For Mixed Type Attributes

- First calculate distance (dissimilarity) matrices for each of the attributes separately
- We have already calculated the dissimilarity measures for Color and Quality
- Now, we will calculate for Quantity

Object	Color (Categorical)	Quality (Ordinal)	Quantity (Numeric)
1	R	Excellent	475
2	B	Fair	10
3	G	Good	1000
4	R	Excellent	500

- For numeric attribute like Quantity, we have to normalize the values so that the values can be mapped [0.0, 1.0]

$$d(i,j) = \frac{x_{if} - x_{jf}}{max - min}$$

Color

$$d = \begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \end{array} \begin{array}{c} \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\ \begin{bmatrix} 0 & & & \\ 1 & 0 & & \\ 1 & 1 & 0 & \\ 0 & 1 & 1 & 0 \end{bmatrix} \end{array}$$

Quality

$$d = \begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \end{array} \begin{array}{c} \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\ \begin{bmatrix} 0 & & & \\ 1.0 & 0 & & \\ 0.5 & 0.5 & 0 & \\ 0 & 1.0 & 0.5 & 0 \end{bmatrix} \end{array}$$

Quantity

$$d = \begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \end{array} \begin{array}{c} \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\ \begin{bmatrix} 0 & & & \\ & 0 & & \\ & & 0 & \\ & & & 0 \end{bmatrix} \end{array}$$

Proximity Measure For Mixed Type Attributes

	$\frac{\lvert x_{if} - x_{jf} \rvert}{max - min}$		$\frac{\lvert x_{if} - x_{jf} \rvert}{max - min}$		$\frac{\lvert x_{if} - x_{jf} \rvert}{max - min}$
d(2,1)	$\frac{\lvert 10 - 475 \rvert}{1000 - 10} = 0.470$				
d(3,1)	$\frac{\lvert 1000 - 475 \rvert}{1000 - 10} = 0.530$	d(3,2)	$\frac{\lvert 1000 - 10 \rvert}{1000 - 10} = 1.0$		
d(4,1)	$\frac{\lvert 500 - 475 \rvert}{1000 - 10} = 0.025$	d(4,2)	$\frac{\lvert 500 - 10 \rvert}{1000 - 10} = 0.4949$	d(4,3)	$\frac{\lvert 500 - 1000 \rvert}{1000 - 10} = 0.505$

Object	Color (Categorical)	Quality (Ordinal)	Quantity (Numeric)
1	R	Excellent	475
2	B	Fair	10
3	G	Good	1000
4	R	Excellent	500

$$d = \begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \end{array} \begin{array}{c} \text{Quantity} \\ \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\ \begin{bmatrix} 0 & & & \\ 0.470 & 0 & & \\ 0.530 & 1.0 & 0 & \\ 0.025 & 0.4949 & 0.505 & 0 \end{bmatrix} \end{array}$$

Proximity Measure For Mixed Type Attributes

Mixed Attributes

$$d(i,j) = \frac{\sum_{f=1}^{p} \delta_{ij}^{(f)} d_{ij}^{(f)}}{\sum_{f=1}^{p} \delta_{ij}^{(f)}}$$

Object	Color (Categorical)	Quality (Ordinal)	Quantity (Numeric)
1	R	Excellent	475
2	B	Fair	10
3	G	Good	1000
4	R	Excellent	500

- $\delta_{ij}^{(f)} = 1$, because
  - None of the attribute values in the objects is missing
  - No attribute has zero value
- Sample calculation for d(2,1)

$\delta_{ij}^{(f)}$

$$d(2,1) = \frac{(1 \times 1) + (1 \times 1.0) + (1 \times 0.470)}{1 + 1 + 1} = \frac{2.470}{3} = 0.823$$

Color

$$d = \begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \end{array} \begin{array}{c} \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\ \begin{bmatrix} 0 & & & \\ 1 & 0 & & \\ 1 & 1 & 0 & \\ 0 & 1 & 1 & 0 \end{bmatrix} \end{array}$$

Quality

$$d = \begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \end{array} \begin{array}{c} \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\ \begin{bmatrix} 0 & & & \\ 1.0 & 0 & & \\ 0.5 & 0.5 & 0 & \\ 0 & 1.0 & 0.5 & 0 \end{bmatrix} \end{array}$$

Quantity

$$d = \begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \end{array} \begin{array}{c} \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\ \begin{bmatrix} 0 & & & \\ 0.470 & 0 & & \\ 0.530 & 1.0 & 0 & \\ 0.025 & 0.4949 & 0.505 & 0 \end{bmatrix} \end{array}$$

Proximity Measure For Mixed Type Attributes

Mixed Attributes

	$\frac{\sum_{f=1}^{p}\delta_{ij}^{(f)}d_{ij}^{(f)}}{\sum_{f=1}^{p}\delta_{ij}^{(f)}}$		$\frac{\sum_{f=1}^{p}\delta_{ij}^{(f)}d_{ij}^{(f)}}{\sum_{f=1}^{p}\delta_{ij}^{(f)}}$
d(2,1)	$\frac{(1\times 1)+(1\times 1)+(1\times 0.470)}{1+1+1}=0.823$		
d(3,1)	$\frac{(1\times 1)+(1\times 0.5)+(1\times 0.530)}{1+1+1}=0.677$	d(3,2)	$\frac{(1\times 1)+(1\times 0.5)+(1\times 1.0)}{1+1+1}=0.833$
d(4,1)	$\frac{(1\times 0)+(1\times 0)+(1\times 0.025)}{1+1+1}=0.0083$	d(4,2)	$\frac{(1\times 1)+(1\times 1)+(1\times 0.4949)}{1+1+1}=0.832$
d(4,3)	$\frac{(1\times 1)+(1\times 0.5)+(1\times 0.505)}{1+1+1}=0.668$		

Color

$$d=\begin{array}{c|cccc} & 1 & 2 & 3 & 4 \\ \hline 1 & 0 & & & \\ 2 & 1 & 0 & & \\ 3 & 1 & 1 & 0 & \\ 4 & 0 & 1 & 1 & 0 \end{array}$$

Quality

$$d=\begin{array}{c|cccc} & 1 & 2 & 3 & 4 \\ \hline 1 & 0 & & & \\ 2 & 1.0 & 0 & & \\ 3 & 0.5 & 0.5 & 0 & \\ 4 & 0 & 1.0 & 0.5 & 0 \end{array}$$

Quantity

$$d=\begin{array}{c|cccc} & 1 & 2 & 3 & 4 \\ \hline 1 & 0 & & & \\ 2 & 0.470 & 0 & & \\ 3 & 0.530 & 1.0 & 0 & \\ 4 & 0.025 & 0.4949 & 0.505 & 0 \end{array}$$

Overall Dissimilarity Matrix

$$d=\begin{array}{c|cccc} & 1 & 2 & 3 & 4 \\ \hline 1 & 0 & & & \\ 2 & 0.823 & 0 & & \\ 3 & 0.677 & 0.833 & 0 & \\ 4 & 0.0083 & 0.832 & 0.668 & 0 \end{array}$$

Cosine Similarity

Cosine Similarity

- A **document** can be represented by thousands of attributes, each recording the *frequency* of a particular word (such as keywords) or phrase in the document.

Document	*team*	*coach*	*hockey*	*baseball*	*soccer*	*penalty*	*score*	*win*	*loss*	*season*
Document1	5	0	3	0	2	0	0	2	0	0
Document2	3	0	2	0	1	1	0	1	0	1
Document3	0	7	0	2	1	0	0	3	0	0
Document4	0	1	0	0	1	2	2	0	3	0

- Other vector objects: gene features in micro-arrays, …
- Applications: information retrieval, biologic taxonomy, gene feature mapping, ...
- Cosine measure: If d_1 and d_2 are two vectors (e.g., term-frequency vectors), then

$$\cos(d_1, d_2) = (d_1 \bullet d_2) / ||d_1|| \; ||d_2|| ,$$

where $\bullet$ indicates vector dot product, $||d||$: the length of vector d

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Example: Cosine Similarity

- $\cos(d_1, d_2) = (d_1 \bullet d_2) / ||d_1|| \; ||d_2||$,
  where $\bullet$ indicates vector dot product, $||d|$: the length of vector d

- Ex: Find the **similarity** between documents 1 and 2.

$d_1 = (5, 0, 3, 0, 2, 0, 0, 2, 0, 0)$
$d_2 = (3, 0, 2, 0, 1, 1, 0, 1, 0, 1)$

$d_1 \bullet d_2 = 5*3+0*0+3*2+0*0+2*1+0*1+0*1+2*1+0*0+0*1 = 25$
$||d_1|| = (5*5+0*0+3*3+0*0+2*2+0*0+0*0+2*2+0*0+0*0)^{0.5} = (42)^{0.5} = 6.481$
$||d_2|| = (3*3+0*0+2*2+0*0+1*1+1*1+0*0+1*1+0*0+1*1)^{0.5} = (17)^{0.5} = 4.12$
$\cos(d_1, d_2) = 0.94$

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Text Books

T1	Introduction to Data Mining, by Tan, Steinbach and Vipin Kumar
T2	Introducing Data Science by Cielen, Meysman and Ali
T3	Storytelling with Data, A data visualization guide for business professionals, by Cole Nussbaumer Knaflic; Wiley
T4	Data Mining: Concepts and Techniques, Third Edition by Jiawei Han and Micheline Kamber Morgan Kaufmann Publishers

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Thank You